

HEALTHCARE DATA SCIENTIST AND  
ANALYST PORTFOLIO

# Dr Mahreen Kiran

Turning complex health data  
into clear, clinically meaningful  
evidence

PhD-trained data scientist with more than four years of experience across population health, clinical prediction, infectious disease analytics and explainable AI. I build reproducible analytical workflows, communicate uncertainty clearly and translate model outputs into decisions that healthcare and public health teams can use.

Contact me

LinkedIn

UK Biobank  
participants  
represented

**500k+** in  
population-  
scale  
research

C-index  
achieved  
in time-  
to-event  
diabetes  
prediction

**0.90**

Patient  
records  
analysed  
in a six-  
year  
dengue  
study

**3,000**

Peer-  
reviewed  
papers  
plus a  
book  
chapter

**5+**

**PhD**

Computer  
Science,  
healthcare-  
focused

**33  
years**

AI and  
diabetes

**1,000**

Monte Carlo  
simulations  
in an

**No  
sponsorship**

Global Talent Visa, available  
for UK roles

predictive  
analytics

literature  
mapped  
through  
bibliometric  
analysis

explainable  
AI study

## WHAT I BRING

# Healthcare analytics from data preparation to decision-ready insight

01

## Health data preparation

Cohort definition, inclusion and exclusion criteria, missing-data assessment, recoding, outlier checks, feature engineering and traceable analytical datasets.

pandas

NumPy

SQL

Excel

02

## Statistical and predictive modelling

Regression, classification, survival analysis, neural networks, ensemble models, causal inference, validation, calibration and robustness testing.

scikit-learn

lifelines

PyTorch

TensorFlow

03

## Insight communication

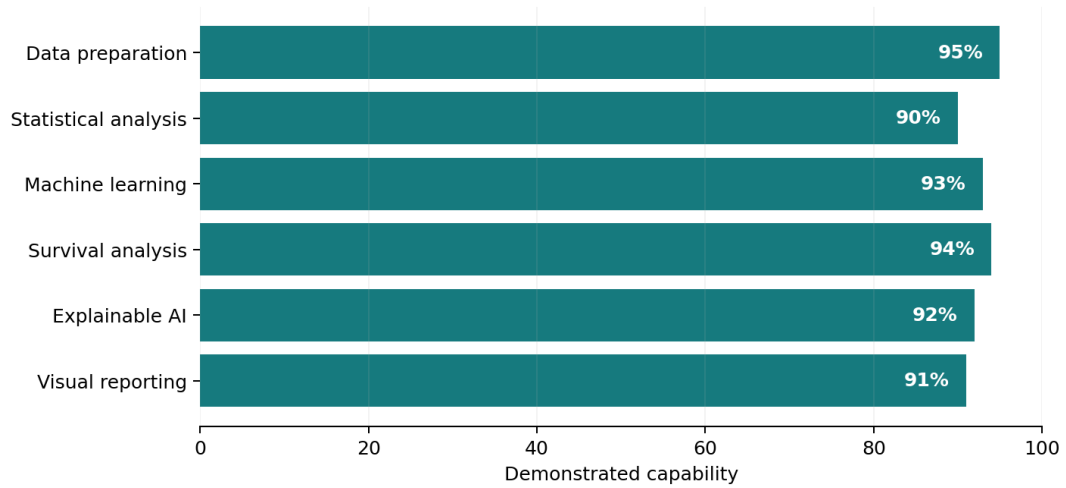
Clear reports, publication-ready visualisations, dashboards and explanations designed for clinicians, researchers, operational teams and senior stakeholders.

Power BI

Matplotlib

SHAP

VOSviewer



## SELECTED WORK

# Projects that demonstrate healthcare impact

Each project below is presented as an employer-facing case study: the question, analytical approach, key result, practical relevance and tools used.

0.90

500k+

18

Figure 1 is a survival plot showing survival probabilities over time (0 to 14 years) for three groups: Very High (25% survival), High (50% survival), and Moderate (75% survival). The plot includes survival curves and a risk score range legend.

**Survival Percentiles**

- 25% Survival (Blue dashed line)
- 50% Survival (Black dashed line)
- 75% Survival (Red dashed line)

**Survival Curves**

- Very High: 25%: 8.0 yrs**
- High: 50%: 8.0 yrs**
- Moderate: 75%: 7.0 yrs**
- Low: 25%: Very High: 50%: 9 yrs**
- High: 50%: 9.0 yrs**
- Moderate: 50%: Very High: 55%: 11.0 yrs**
- Low: 50%: High: 75%: 12.0 yrs**
- Very Low: 25%: 13.0 yrs**
- Moderate: 75%: 14.0 yrs**

**Risk Score Range**

- Y Low: [-1.47 - 0.22]
- Y: [0.23 - 0.40]
- Y High: [0.41 - 0.70]
- Y High: [1.12 - 2.75]

: Survival probability is plotted across time for five strata: very low, low, moderate, and high (defined by risk score ranges). Shaded bands indicate confidence intervals. Vertical lines indicate 25%, 50%, and 75% survival percentiles, with labels showing estimated survival probabilities.

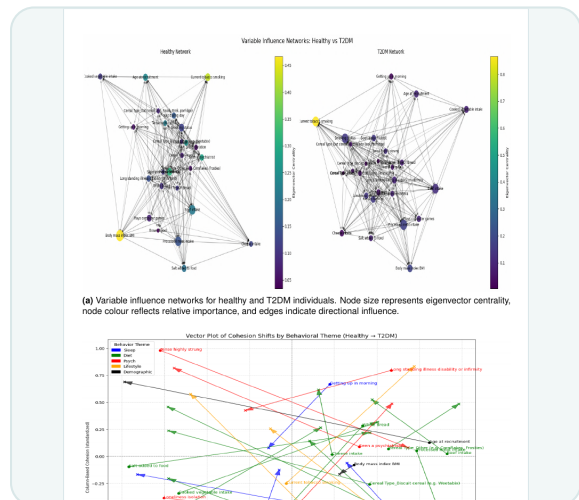
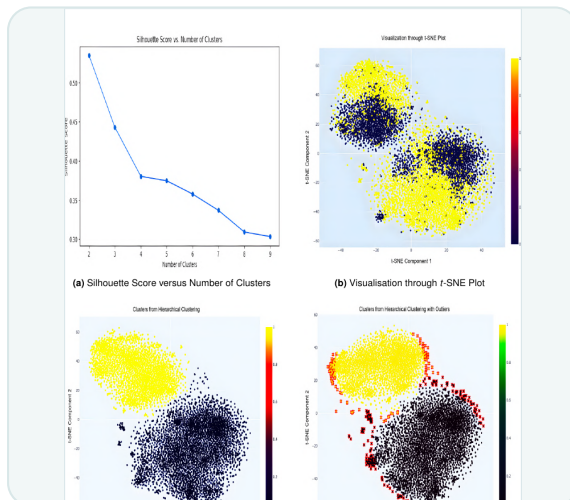
Demonstrates experience with longitudinal health data, clinical risk stratification, model validation, interpretable outputs and responsible communication of modifiable risk factors.

## What I did

- Defined analytic cohorts and prepared behavioural, lifestyle, psychosocial and health variables.
- Built penalised Cox models and neural approaches for time-to-event risk estimation.
- Used causal reasoning and intervention simulation to examine potential changes in sleep, diet and mental health.
- Created risk groups and survival trajectories to make model results easier to interpret.

## Key findings and tools

- Loneliness, insomnia, mental health history, smoking and dietary behaviours contributed meaningful risk information.
- Healthy sleep duration showed a protective association, while several psychosocial stressors increased estimated hazard.
- **Tools:** Python, pandas, NumPy, lifelines, scikit-learn, neural networks, SHAP, causal inference, Matplotlib.



RESEARCH COLLABORATION |  
INFECTIOUS DISEASE  
ANALYTICS

## Machine learning prediction of dengue infection across five districts

**Challenge:** Analyse a complex, imbalanced, multi-year patient dataset that combined clinical, laboratory, serological, behavioural, epidemiological and environmental information, then determine what genuinely contributed to case classification.

**3,000**

Patient  
records

**2019–  
24**

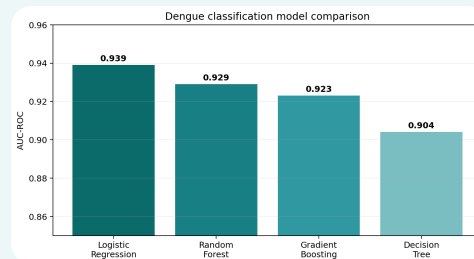
Six-year  
study  
period

**5**

Endemic  
districts

**Strong analytical judgement:**

The analysis did not simply celebrate a high score. It identified that serological variables drove much of the full-model performance and showed that environmental variables alone were near chance for individual-level confirmation. This distinction



protects against misleading claims.

### Analytical approach

- Data-quality checks and preprocessing across more than 50 variables.
- Class imbalance handled within training folds using SMOTE and class-weighted learning.
- Compared Logistic Regression, Random Forest, Gradient Boosting and Decision Tree models.
- Used stratified cross-validation, temporal and geographic validation, feature importance and ablation analysis.

### Outcome and relevance

- Initial model comparisons produced AUC-ROC values around 0.90 to 0.94.
- A later full-feature model reached 0.995 AUC, but sensitivity analysis showed this was substantially driven by diagnostic serology.
- Environmental-only AUC remained approximately 0.49 to 0.52, supporting their use at population or temporal surveillance level rather than individual diagnosis.
- **Tools:** Python, scikit-learn, pandas, cross-validation, SMOTE, feature importance, ablation analysis.



# Climate-aware prediction of fungicide effectiveness using XGBoost and SHAP

**Challenge:** Integrate field trials, laboratory results, climate variables and molecular modelling to understand how temperature, humidity and precipitation altered pesticide effectiveness.

1,000

Monte Carlo simulations

3

Key esterase proteins identified

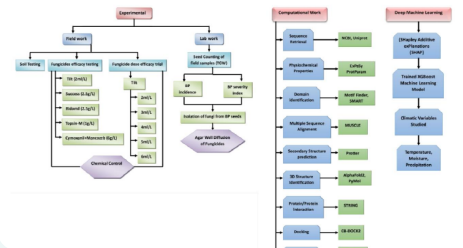
-5.1

Best reported docking score

**Transferable value:** Although this project is agricultural, it directly demonstrates skills relevant to healthcare analytics: multimodal data integration, interaction effects, explainability, simulation, uncertainty and translation of technical outputs into practical recommendations.

## Material and Methods

This study's methodology consists of four primary components: fieldwork, laboratory work, computational analysis, and machine learning employing XGBoost (Extreme Gradient Boosting) in conjunction with Explainable AI (XAI) techniques, particularly utilizing SHAP (SHapley Additive exPlanations) to deliver interpretable and transparent insights regarding feature interactions and their influence on pesticide efficacy (Prendin et al., 2023).



### What I contributed

- Prepared climate and pesticide variables, including imputation, scaling, categorical encoding and outlier checks.
- Applied XGBoost and SHAP interaction analysis to quantify climate-dependent pesticide behaviour.
- Used simulation to test combined environmental conditions and improve interpretability.

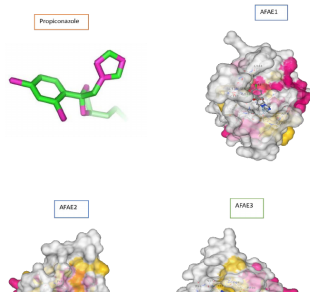
### Key result

- Propiconazole performed strongly, with Dose 4 producing the best disease suppression and yield improvement.
- SHAP results separated moisture-dependent, region-specific and climate-adaptable pesticides.
- **Tools:** XGBoost, SHAP, Monte Carlo simulation, AlphaFold 2, molecular docking, Python.

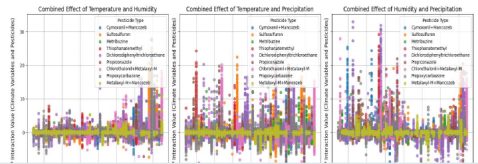
to help with ligand binding. Various fungicides were tested as ligands for docking with the Esterase\_PHD of *A. blightis*. Propiconazole (Tili) was found to disrupt ergosterol biosynthesis, potentially affecting ester bond hydrolysis. Chlorothalonil (Cymoxanil) interfered with multiple fungal processes, potentially hindering esterase function. Additionally, Cymoxanil inhibited enzyme activity by targeting cytochrome C oxidase. These interactions were visually represented, offering insights into their mechanisms within the Esterase\_phd domain (Table S17).

#### Docking

The Esterase\_phd domain (depicted in white and pink as a cartoon) interacted with fungicides shown in space-fill style (Figure 3), revealing specific interactions. For instance, Propiconazole was bound to amino acid residues in the C1 pocket of proteins AFAE1 (5.1), AFAE2 (5.1), and AFAE3 (-4.7). It formed bonds with multiple residues in the C1 pocket across all three proteins, indicating strong binding. Similarly, Cymoxanil bound to specific residues in the C1 pocket of all three proteins, achieving scores of 5.1 (AFAE1, AFAE2) and 4.6 (AFAE3). These docking simulations provided insights into the binding affinities and interactions of fungicides with the Esterase\_phd domain, guiding research for therapeutic agents targeting these proteins for various applications, including disease management and drug discovery (Table S18).



in response to changing weather patterns. Propoxycarbazon and Metribuzine showed their most effectiveness in balanced conditions with moderate temperature, humidity, and precipitation, whereas Thiophanate-methyl and Sulfosulfuron functioned consistently in semi-arid areas with moderate precipitation and reduced humidity. These results underlined how crucial it is to match pesticide application methods to local climate circumstances to maximize crop protection, minimize waste, and improve sustainability. It was advised to prioritize moisture-dependent pesticides like Propiconazole or Cymoxanil + Mancozeb in wet conditions, while Thiophanate-methyl or Sulfosulfuron produced more dependable outcomes in dry areas. To increase agricultural productivity, this study laid the groundwork for climate-smart pest management techniques (Figure 4).



# Automated tuberculosis detection from chest radiographs

**Challenge:** Work with heterogeneous chest X-ray data collected from multiple hospitals and design a reproducible image-processing pipeline capable of highlighting pathological regions associated with tuberculosis.

## Multi-site

Hospital  
imaging  
sources

## End-to-end

Preprocessing  
to  
classification

## Clinical AI

Diagnostic  
support  
context



## Chest X-ray analytics

Image curation,  
preprocessing,  
segmentation and  
classical machine learning

### Work completed

- Collected and curated X-ray datasets with variable formats and quality.
- Designed preprocessing and segmentation workflows to isolate clinically relevant regions.
- Applied computer vision and machine learning methods to

### Transferable strengths

- Clinical imaging data quality and standardisation.
- Feature extraction and segmentation.
- Understanding of diagnostic support rather than autonomous decision-making.

support automated diagnostic analysis.

## PROFESSIONAL BACKGROUND

# Research, teaching and multidisciplinary collaboration

I combine technical depth with the ability to explain analytical choices to people from clinical, scientific, operational and academic backgrounds.

**2022–2026**

### **Data Scientist, PhD Research | Anglia Ruskin University**

Population-scale diabetes prediction, survival analysis, Digital Twin simulation, behavioural network analysis, explainability, causal inference and publication.

**2025–2026**

### **Research Associate | Cranfield University**

Analysed real-world maritime energy data and translated operational findings into recommendations for engineering and sustainability stakeholders.

**Sep 2024**

### **Data Study Group Participant | Alan Turing Institute and UCL Hospital**

Worked in a multidisciplinary team on a clinical prediction challenge covering preprocessing, statistical analysis, model development and reporting.

**2013–2022**

## Lecturer in Computer Science

Taught databases, image processing, programming and data structures, supervised research projects and communicated complex methods to mixed-ability audiences.

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### RESEARCH COMMUNICATION

## Selected publications and evidence outputs

Peer-reviewed work demonstrates the ability to document methods, defend analytical decisions and communicate findings clearly.

### **A Digital Twin Framework for Predicting and Simulating Type 2 Diabetes Onset**

Frontiers in Digital Health, 2026

### **Type 2 Diabetes Prediction Without Labs: A Systems-Level Neural Framework**

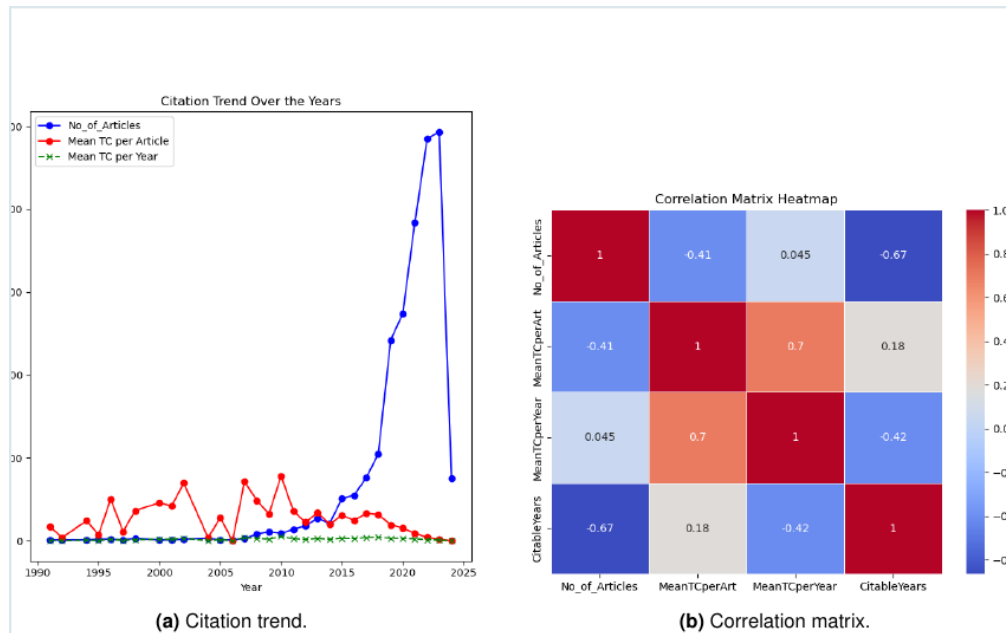
Frontiers in Digital Health, 2026

### **Machine Learning and Artificial Intelligence in Type 2 Diabetes Prediction**

33-year bibliometric and literature analysis, 2025

### **Explainable AI and Machine Learning for Wheat Black Point Disease**

Multidisciplinary preprint, 2025



**Figure 3.2:** Analysis of research articles published annually with respect to citations. (a) Citation trend showing the growth in the number of published articles, mean citations per article, and mean citations per year from 1990 to 2024. (b) Correlation matrix heatmap illustrating relationships among article count, citation averages, and citable years, where darker red indicates stronger positive correlation.

OPEN TO HEALTHCARE ANALYTICS  
OPPORTUNITIES

Let's turn  
health data  
into evidence  
that improves  
decisions.

Target roles: Healthcare Data Analyst,  
Senior Data Analyst, Clinical Data Analyst,

[mehreen.kiran89@gmail.com](mailto:mehreen.kiran89@gmail.com)

+44 7473 949490

[linkedin.com/in/mahreen-kiran](https://www.linkedin.com/in/mahreen-kiran)

Portsmouth, UK

Population Health Analyst and Health Data  
Scientist.

Portfolio prepared from Dr Mahreen Kiran's CV, doctoral thesis and research manuscripts.  
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